1.8 Reaction Kinetics

Question Paper

Course	CIE A Level Chemistry
Section	1. Physical Chemistry
Topic	1.8 Reaction Kinetics
Difficulty	Hard

Time allowed: 50

Score: /41

Percentage: /100

Question 1a

Reaction rates can be affected by a range of factors including changes in pressure and temperature.

Fig 1.1

On Fig 1.1:

i)
Sketch one Maxwell-Boltzmann distribution labelled T_1 and a second Maxwell-Boltzmann distribution at a higher temperature labelled T_2.

[2]

ii)
State how the mean energy of the molecules would be at T_2 compared to T_1.

[2]

[4 marks]

Question 1b

Hydrogen iodide can be used in the manufacturing of pharmaceuticals and can be broken down back into its elements in standard form, iodine and hydrogen.

$2HI\,(g) \rightarrow H_2(g) + I_2(g)\ \Delta H = -52\ kJ\ mol^{-1}$

The activation energy when uncatalysed is +183 kJ mol^{-1} and when catalysed with gold it is +105 kJ mol^{-1}.

i)
Sketch a reaction profile for the reaction, including the curves for the activation energies for both the catalysed and uncatalysed reactions.

ii)
Calculate the activation energy for the *reverse reaction* in both the uncatalysed and catalysed reactions.

iii)
Explain why increasing the concentration of hydrogen iodide gas results in a faster reaction rate.

[6 marks]

Question 1c

Catalysts are often used in industrial processes and can be used in a variety of forms.

i)
Explain why it is likely that the solid gold catalyst was used in powder form to catalyse the reaction mentioned in part (c).

[1]

ii)
Gold is a heterogeneous catalyst used in the formation of hydrogen iodide. State the difference between a homogenous and heterogenous catalyst.

[1]

iii)
State how, if at all, the area under the curve of a Maxwell-Boltzmann distribution curve, changes as a catalyst is introduced without changing the temperature or the total number of molecules.

[1]

[3 marks]

Question 1d

The Contact process is an important industrial process, contributing to the production of sulfuric acid. In the Contact process, solid vanadium (V) oxide, a heterogeneous catalyst, is used to make sulfur trioxide from sulfur dioxide and oxygen. This process is reversible.

i)
Write a balanced symbol equation for this reaction. Include state symbols in your answer.

[1]

ii)
Explain why the use of the catalyst in the Contact process, reduces energy demand and benefits the environment.

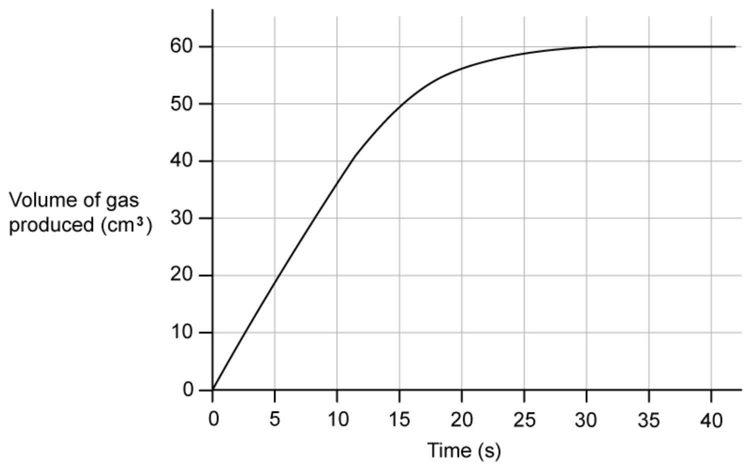
[2]

[3 marks]

Question 2a

0.5 g of magnesium reacts with 50 cm^3 of 0.01 $mol dm^{-3}$ nitric acid. Magnesium is in excess.

A graph monitoring the volume of hydrogen gas produced is shown below:



i)
Calculate the mean rate of reaction over the first 15 seconds of the reaction

[1]

ii)
Calculate the actual rate of reaction at 15 seconds

[3]

iii)
Explain the difference in values for rate

[1]

[5 marks]

Question 2b

Compare the expected rate and progress of the reaction if 25 cm^3 of 0.2 mol dm^{-3} nitric acid was used instead of 50 cm^3 of 0.1 mol dm^{-3} nitric acid.

[3 marks]

Question 2c

Suggest one change to the reaction that could be made to produce more hydrogen gas in total and explain your choice.

[2 marks]

Question 2d

Suggest why it is often better to study a slower reaction instead of a faster one.

[2 marks]

Question 3a

A Maxwell-Boltzmann distribution curve is shown below in Fig 3.1

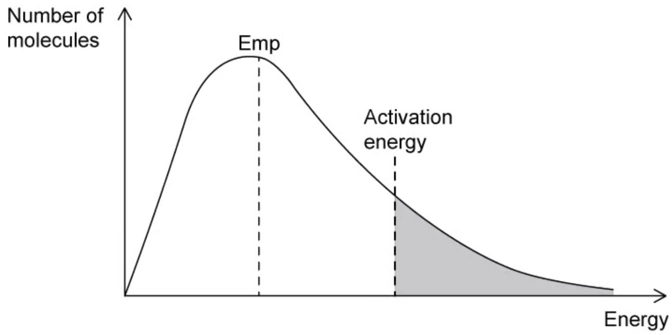


Fig 3.1

For the changes detailed in part **(i)** and **(ii)** state and explain the effect the change would have on:

- The area under the curve
- The value of the most probable energy of the molecules, E_{mp}
- The proportion of molecules with greater than or equal to E_a

i)

The temperature of the original reaction is increased, but no other changes are made.

[2]

ii)

A catalyst is added to the original reaction mixture, but no other changes are made.

[2]

[4 marks]

Question 3b

A chemist performed a reaction at three different temperatures, 100 K, 300 K and 700 K as shown by the Maxwell-Boltzmann distribution graph in Fig 3.2.

Fig 3.2

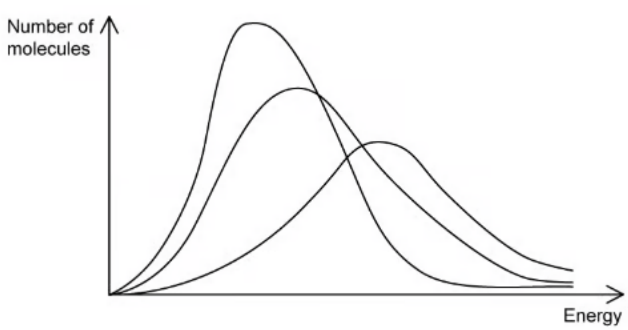


Fig 3.2

i)
Label each curve in Fig 3.2 ith the correct temperature values, 100 K, 300 K and 700 K.

[1]

ii)
Consider the following statement, 'All reacting molecules have higher kinetic energy at 700 K than they do at 300 K'.

State whether you agree this statement is correct and justify your reasons.

[3]

[4 marks]

Question 3c

Hydrogen will react with chlorine to form the hydrogen halide, hydrogen chloride, a colourless gas.

$$H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$$

i)

Give one reason why most collisions between hydrogen and chlorine molecules do not lead to the formation of hydrogen chloride.

[1]

ii)

Apart from changing the temperature, state and explain two ways of speeding up the formation of hydrogen chloride.

[4]

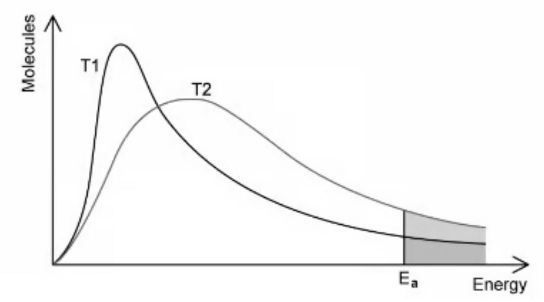
[5 marks]

Model Answers: Hard

1a

a)

i) A labelled Maxwell-Boltzmann curve for T_1 and T_2 would look like:



- Axes correctly labelled with (number of) molecules and (kinetic) energy; [1 mark]
- Both curves start at the first small square nearest the x axis
  AND
  Both lines must not go below the x axis; [1 mark]
- Two lines clearly labelled (T_1) and (T_2) with T_2 being to the right and having a lower peak than T_1, T_2 only crosses T_1 once and drawn above T_1 at the higher energy; [1 mark]

ii) At T_2 the mean energy of the molecules will be:

- Higher / bigger / greater / increase; [1 mark]

[Total: 4 marks]

- **Careful**: Maxwell-Boltzmann curves
  - Must start at the origin (as no molecules have some energy)
  - Curves do not meet the x axis (as there is no maximum energy for a molecule)
- You are asked for a labelled sketch - so make sure you don't forget to label your axes, activation energy (Ea) and clearly label the two different temperature curves (T_1 and T_2)
- A higher proportion of molecules have energy ≥ the E_a, resulting in more frequent

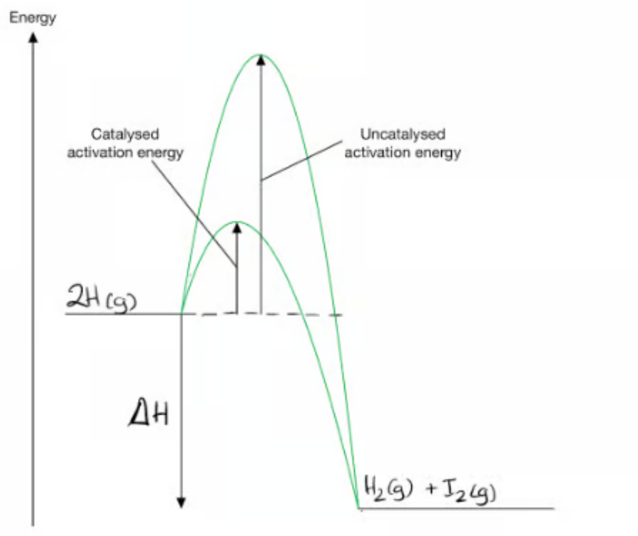
successful / effective collisions at T_2

- If you are asked to explain the effect of temperature avoid stating simply 'more collisions' or 'molecules have gained more energy' or 'molecules have energetic collisions' as they do not fully explain the kinetics behind raising the temperature

1b

b)

i) The energy profile diagram for the reaction is:



- An exothermic energy profile diagram has been drawn with fully labelled reactants at a higher energy than products; [1 mark]
- The uncatalysed activation energy curve is higher than catalysed
  AND
  Both curves start at the reactants and end at the products; [1 mark]

ii) The activation energy for the reverse reaction is:

- *Ea* uncatalysed = 52 + 183 = (+) 235 kJ mol^{-1} ; [1 mark]
- *Ea* catalysed = 52 + 105 = (+) 157 kJ mol^{-1} ; [1 mark]

iii) Increasing concentration increases the rate of reaction because:

- More particles / molecules / reactants / moles in the same volume / space
  OR
  Particles / molecules / reactants / moles are closer together; [1 mark]

- Successful collisions are more frequent; [1 mark]

[Total: 6 marks]

- When drawing activation energy curves on an energy profile diagram, make sure that the catalysed and uncatalysed reaction curves do not meet before the products as this can lose a mark
- Calculating the *E*a for the reverse reaction involves going backwards from products to reactants:
  - - 52 kJ mol^{-1} changes to a positive value 52 kJ mol^{-1} as essentially you are reversing the reaction going from exothermic to endothermic, so the enthalpy value also changes sign

1c

c)

i) Gold was used in powdered form because:

- It increases the surface area / provides a larger surface for reactions; [1 mark

ii) The difference between a homogeneous and heterogeneous catalyst is:

- A homogeneous catalyst has the same physical state as the reactants
  AND
  A heterogeneous catalyst has a different physical state from the reactants; [1 mark]

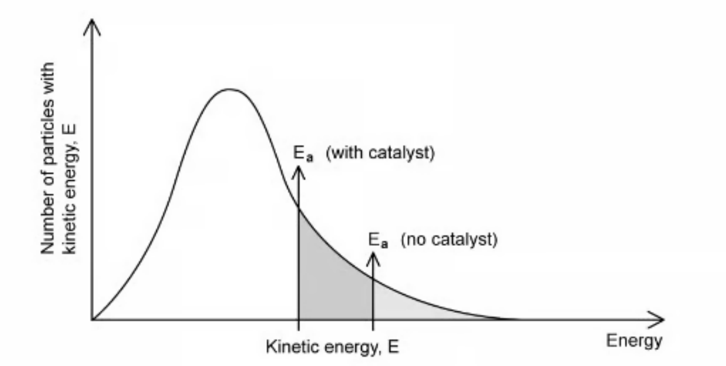
iii) When using a catalyst the area underneath the Maxwell-Boltzmann distribution curve:

- Stays the same; [1 mark]

[Total: 3 marks]

- Many solid catalysts are often used in either a **powdered form** or are part of a **'honeycomb' structure** (e.g. catalysts in a car catalytic converter) to maximise the available surface area for a reaction to happen on:
  - Reactant molecules are **adsorbed** (weakly bonded) onto the surface
  - After the reaction, the products leave the surface of the catalyst by **desorption** (you learn more about this process in the Transition Metal topic at A Level)

- A catalyst works by providing an alternative reaction route, with a lower Ea
    - This means that a greater proportion of molecules have energy above the new Ea, resulting in more particles / molecules colliding with sufficient energy to react:



- So, the curve doesn't shift or change shape (like with temperature), meaning the area under the curve stays the same for a catalyst

1d

d)

i) A balanced symbol equation with state symbols for the contact process is:

- $2\ SO_2(g) + O_2(g) \rightleftharpoons 2\ SO_3(g)$; [1 mark]

ii) A catalyst reduces energy demand and benefits the environment because:

Any **two** from the following:

- Lower temperatures / less heat / less thermal energy is required; [1 mark]
- Less fossil fuels / oil / coal / gas / non-renewable fuels are required; [1 mark]
- CO_2 emissions are reduced / decrease; [1 mark]

[Total: 3 marks]

- Writing the balanced equation involves you picking out key information from the question:
    - Reversible reaction
    - Vanadium (V) oxide is a solid heterogeneous catalyst (meaning the reactants are in a different phase and in this case are gases)
    - You should be able to deduce the formulas of sulfur dioxide and sulfur trioxide from your earlier work on oxidation numbers and formulas
- Do not mention costs or economic advantages for your answer to part (ii)
    - The question asks about how a catalyst reduces energy demands and benefits the environment instead:
    - Making product faster and using less energy can cut costs and increase profitability however that is not the focus of the question in this case
- Don’t be tempted to say it creates less CO_2, less global warming or less greenhouse gases for your answer to part (ii)
- Remember a catalyst works by **affecting the rate at which a product is made**, but **does not affect the amount of product produced**

2a

a)

i) The mean rate of reaction over the first part of the experiment is:

- Rate = $\frac{change\ in\ volume}{change\ in\ time} = \frac{50\ cm^3}{15\ s} = 3.3\ (cm^3\ s^{-1})$; [1]

- Do not mention costs or economic advantages for your answer to part (ii)
    - The question asks about how a catalyst reduces energy demands and benefits the environment instead:
    - Making product faster and using less energy can cut costs and increase profitability however that is not the focus of the question in this case
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- Remember a catalyst works by **affecting the rate at which a product is made**, but **does not affect the amount of product produced**

2a

a)

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ii) The actual rate at 15 s is:

- Tangent drawn on the graph at 15 s; [1]
- Gradient of the tangent used to calculate rate; [1]
- Rate of 1.5 to 2.0 ($cm^3\ s^{-1}$); [1]

iii) The difference in the values is because:

- The fastest rate is at the start of the reaction/ the reaction slows down by 15 s; [1]
  OR
  The graph has started to curve by 15 s/ the gradient is shallower at 15 s; [1]

[Total: 5 marks]

- The mean rate of a reaction is an appropriate calculation for the proportion of the initial reaction where the graph is essentially a straight line
- After this, the rate is best calculated by drawing a tangent to the curve at the point of the given time, and taking the gradient of that tangent. The larger the line drawn, the smaller the uncertainty of the measurement, so draw big tangents!
- Gradients are always calculated by the change in the value on the y-axis divided by the change in value on the x-axis
- A steeper line should give a larger value for the gradient, which represents a faster rate of reaction

2b

b) Using 25 cm^3 of 0.2 mol dm^{-3} nitric acid would mean that:

- The initial rate of reaction would increase/ the initial gradient of the graph would increase; [1]
- The end of the reaction would be reached sooner/ OWTTE; [1]
- The final/ total volume of gas produced would remain at 60 cm^3/ the same; [1]

[Total: 3 marks]

- The two solutions here would contain the same number of moles of acid, which is the limiting reagent since part a) states that the magnesium is in excess, so the final volume of gas produced would be the same in both cases
- The more concentrated acid would react more quickly due to more frequent collisions occurring, and the reaction would end faster due to the acid being used up more quickly

2c

c) Changes to the reaction that would produce more hydrogen gas are:

- Increase the concentration of the acid without decreasing the volume;

OR
Increase the volume of the acid without decreasing the concentration;
OR
Increase the number of moles of acid reacting; [1]
- Increasing the moles/ particles of the limiting reagent increases the amount/ moles/ volume of the product; [1]

[Total: 2 marks]

- The only way to increase the amount (volume/ mass/ moles) of a product is to increase the moles of the limiting reagent being used
- In this case the limiting reagent is an acid and therefore volume or concentration of the acid can be changed

2d

d) It is often better to study a slower reaction as:

Any **two** from the following:

- Shorter reaction times give a larger (%) error in timing; [1]
- It is possible to compare reactions from the initial rate only (so the reaction does not need to always be complete); [1]
- More readings can be taken in the initial part of the reaction, reducing the change of anomalies being missed/ OWTTE; [1]

[Total: 2 marks]

- Reactions that are very fast may seem time efficient and more exciting, but the increased error associated with taking the recordings often makes the calculated rates less accurate
- The initial section of the reaction is all that is needed when comparing rates; an example of this is when clock reactions are conducted. The colour change occurs in the first 10–20% of the reaction and is still suitable to accurately compare rates across reactions

3a

a) For each of the following changes, the effect on the area under the curve, the value of the E_{mp} and the proportion of molecules with energy greater than or equal to E_a is:

i) The temperature of the original reaction is increased, but no other changes are made:

- Area under the curve is unchanged (as number of molecules is unchanged); [1 mark]
- E_{mp} value increases / position of E_{mp} on the curve shifts right / most probable energy of the molecules increases
  AND
  Proportion of molecules with energy greater than or equal to E_a increases; [1 mark]

ii) A catalyst is added to the original reaction mixture:

- Area under the curve is unchanged (as number of molecules is unchanged); [1 mark]
- E_{mp} value does not change / position of E_{mp} on the curve does not change / most probable energy of the molecules does not change
  AND
  Proportion of molecules with energy greater than or equal to E_a increases (as the activation energy has been lowered by the addition of a catalyst); [1 mark]

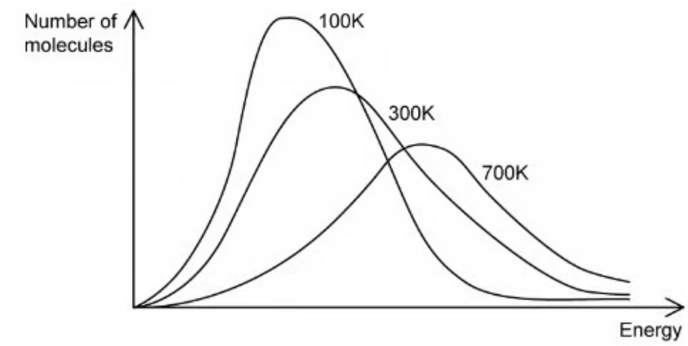
[Total: 6 marks]

- A catalyst lowers the activation energy (offers a different reaction pathway of lower activation energy) meaning more particles will have energy greater than or equal to the activation energy but the most probably energy of the molecules increases

3b

b)

i) The labelled Maxwell-Boltzmann distribution curve:



- Temperatures labelled correctly going from low to high values (left to right); [1 mark]

ii) The correct answer to the statement is:

- The given statement is incorrect; [1 mark]
- The distribution curves for 700 K and 300 K show that there are molecules with both high and low energy values; [1 mark]
- (But) The most probable energy value / E_{mp} will be greater at 700 K than 300 K
  OR
  The peak of the curve is at a higher energy level at 700 K than 300 K; [1 mark]

[Total: 4 marks]

- When you increase the temperature, you are shifting the curve to the right on a Maxwell-Boltzmann distribution graph:
  - The most probable energy of the molecules increases
  - Area under the curve is unchanged (as number of molecules is unchanged)
  - Proportion of molecules with energy greater than or equal to E_a increases

3c

c)

i) The reason why most collisions between hydrogen and chlorine molecules do not lead to the formation of hydrogen chloride is:

- The molecules / particles / reactants don't have enough (kinetic) energy
  OR
  The orientation of the molecules / particles / reactants is wrong; [1 mark]

ii) Two ways to speed up the formation of hydrogen chloride are:

- Increase pressure / reduce the volume
  OR
  Increase concentration; [1 mark]

- This increases the frequency of successful collisions / number of successful collisions per second; [1 mark]
- Use a catalyst; [1 mark]
- This lowers the activation energy / E_a (by providing an alternative pathway); [1 mark]

[Total: 5 marks]

- For a reaction to occur, reactants must have the **correct orientation** and **collide with a minimum collision energy** (the sum of the kinetic energies of the colliding particles) to break bonds in the reactants